Diffusion Model

Q1. Explain the working principle and training procedure of a Denoising Diffusion Probabilistic Model (DDPM). Derive the simplified training objective used in DDPM.

Consider a Denoising Diffusion Probabilistic Model (DDPM) with the following parameters:

- Original data vector: $x_0 = [0.5, -0.2, 0.8]$
- Cumulative noise coefficient: $\bar{\alpha}_t = 0.64$
- Actual noise vector: $\varepsilon = [0.2, -0.4, 0.6]$
- Predicted noise vector: $\varepsilon_\theta = [0.1, -0.1, 0.3]$

(a) Calculate the noisy data vector x_t using the forward diffusion process.
(b) Compute the noise prediction loss using the squared L2 norm.
(c) Estimate the original data vector $\hat{x}_0$ using the predicted noise.
(d) Suppose the updated predicted noise vector is $\varepsilon_\theta' = [0.18, -0.35, 0.55]$. Calculate the updated noise prediction loss and the percentage reduction in loss.
(e) Calculate the reconstructed data vector using the updated noise prediction and compare its reconstruction error with the initial reconstruction.

Q2. Explain the forward diffusion process of DDPM in detail. Derive the diffusion kernel that directly relates a noisy sample x_t to the original data sample x_0. Explain the role of the noise schedule and timestep selection during training.

A Denoising Diffusion Probabilistic Model (DDPM) gradually corrupts an original image by adding Gaussian noise over multiple timesteps. Consider an image pixel with an initial intensity of $x_0 = 0.8$. The forward diffusion process is defined as:

$$x_t = \sqrt{(1 - \beta_t)}\, x_{t-1} + \sqrt{(\beta_t)}\, \varepsilon_t$$

The noise schedule and sampled Gaussian noise values are given below:

Timestep	β_t	ε_t
1	0.1	0.2
2	0.2	−0.3
3	0.3	0.5

Calculate the following:

a) Compute the noisy image values x_1, x_2, and x_3 using the sequential forward diffusion process. Show all intermediate calculations.
b) Calculate the cumulative signal coefficient at each timestep using: $\bar{\alpha}_t = \prod_{s=1}^{t} (1 - \beta_s)$
c) Using the diffusion kernel, directly calculate x_3 from x_0 and the combined Gaussian noise. Compare this result with the sequential calculation in part (a).
d) Based on the results, explain how increasing the diffusion timestep affects the original signal and noise contributions. Why does the noisy pixel intensity not necessarily decrease monotonically?

Q3. Describe the reverse denoising process of a diffusion model. Explain how a neural network estimates the noise at each timestep. Why is the reverse conditional distribution generally difficult to compute directly, and how is it approximated?

A DDPM performs reverse denoising at timestep \(t\) using the following equation:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left[x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right] + \sigma_t z$$

Consider the following parameters:

- Current noisy sample: $x_t = 0.75$
- Noise schedule parameter: $\alpha_t = 0.8$
- Noise variance: $\beta_t = 0.2$
- Cumulative signal coefficient: $\bar{\alpha}_t = 0.5$
- Predicted noise: $\varepsilon_\theta(x_t,t) = 0.6$
- Reverse-process standard deviation: $\sigma_t = 0.15$
- Random Gaussian noise: $z = -0.4$

Calculate:

a) Calculate the noise correction term.
b) Determine the estimated reverse diffusion value x_{t-1}.
c) Calculate x_{t-1} when the stochastic noise term is removed.
d) Explain the role of the stochastic component in the reverse denoising process.

Q4. Explain the architecture of U-Net used in diffusion models for noise prediction. Discuss the functions of its major components

Consider a diffusion model employs a U-Net architecture to predict the noise added to an input image during the forward diffusion process. Consider a simplified U-Net with three encoder blocks, a bottleneck, and three decoder blocks. The feature map dimensions are given below.

Layer	**Feature map dimensions**
Input image	64 × 64 × 3
Encoder Block 1	32 × 32 × 64
Encoder Block 2	16 × 16 × 128
Encoder Block 3	8 × 8 × 256
Bottleneck	4 × 4 × 512
Decoder Block 1	8 × 8 × 256
Decoder Block 2	16 × 16 × 128
Decoder Block 3	32 × 32 × 64
Output layer	64 × 64 × 3

Assume that each decoder block concatenates its feature map with the corresponding encoder feature map through skip connections. Calculate the following:

a) Calculate the total number of feature elements in Encoder Block 2 and Bottleneck Block.
b) During the first decoder stage, the upsampled bottleneck feature map has dimensions 8 × 8 × 256. It is concatenated with the Encoder Block 3 feature map of dimensions 8 × 8 × 256. Determine the dimensions of the resulting feature map after concatenation.
c) Calculate the total number of feature elements in the resulting feature map obtained in part (b). Compare it with the feature map size before concatenation.
d) If a residual block receives an input feature vector of [0.2, 0.4, 0.6] and the output of its convolutional transformation is [0.1, −0.2, 0.3], calculate the output of the residual connection.
e) The actual noise added to an image is ε =[0.2, 0.5, -0.3, 0.1], whereas the U-Net predicts ε_θ =[0.1, 0.4, -0.1, 0.2]. Calculate the mean squared error (MSE) between the actual and predicted noise.
f) Explain how encoder blocks, decoder blocks, skip connections, and residual blocks contribute to noise prediction in diffusion models.

Q5. Explain the concept and working principle of Denoising Score Matching (DSM).

Consider a one-dimensional noisy data sample with x_t=2.5. The predicted score by a neural network is $s_\theta(x_t,t)$=-1.2, while the target score is $s^*(x_t,t)$=-1.5. The Denoising Score Matching loss is defined as: $L = ||s_\theta(x_t, t) - s^*(x_t, t)||_2^2$. Answer the following:

a) Define the score function mathematically and calculate the score prediction error.
b) Calculate the Denoising Score Matching loss for the given sample.
c) If the predicted score is updated to -1.45, calculate the new loss and percentage reduction in loss.
d) Using the relationship between score estimation and noise prediction ($\varepsilon_\theta = - \sigma\, s_\theta(x_t, t)$), Calculate the actual and predicted noise values before and after the parameter update, assuming the noise standard deviation is σ =0.5.
e) Explain how minimizing the score matching loss helps a diffusion model learn the reverse denoising process.

Q6. Explain Denoising Diffusion Implicit Models (DDIM) and their underlying non-Markovian diffusion formulation. Discuss the significance of the parameter σ_t and explain the deterministic sampling process when σ_0.

Consider A pretrained diffusion model is used to generate high-resolution images. The DDPM sampler requires 1,000 denoising steps, whereas the DDIM sampler generates images using 50 steps. Assume that each U-Net evaluation requires 25 milliseconds and that the computational time for other operations is negligible. Calculate:

a) The total sampling time required by DDPM.
b) The total sampling time required by DDIM.
c) Calculate the percentage reduction in sampling time using DDIM.
d) If 200 images are generated, calculate the total computational time saved by using DDIM instead of DDPM.

Q7. Explain the working principles of classifier guidance and classifier-free guidance in conditional diffusion models. How is classifier information incorporated into the reverse denoising process?

Consider A conditional diffusion model uses classifier-free guidance to control the image generation process. The guided score function is defined as:

$$s_{guided} = s_{unconditional} + \omega(s_{conditional} - s_{unconditional})$$

Consider the following predicted score vectors:

$$\text{Unconditional score: } s_{unconditional} = [0.2, 0.4, 0.6]$$
$$\text{Conditional score: } s_{conditional} = [0.8, 0.6, 0.2]$$

Calculate:

a) Determine the guided score vector for guidance scale $\omega = 1$.
b) Determine the guided score vector for guidance scale $\omega = 2$.
c) Determine the guided score vector for guidance scale $\omega = 3$.
d) Calculate the Euclidean distance between the guided score and unconditional score for each guidance scale.
e) Explain how increasing the guidance scale affects the quality and diversity of generated samples.

Q8. Explain the architecture and working principle of latent diffusion models. Discuss how a Variational Autoencoder (VAE) is combined with a diffusion model to perform the diffusion process in a compressed latent space. What are the major advantages of latent diffusion over pixel-space diffusion?

Consider A Variational Autoencoder (VAE) encoder compresses an input image of dimensions $256 \times 256 \times 3$ into a latent representation of dimensions $32 \times 32 \times 4$. A diffusion model is trained either in pixel space or in latent space. Calculate:

a) Determine the total number of elements in the original image.
b) Determine the total number of elements in the latent representation.
c) Calculate the compression ratio achieved by the VAE.
d) Calculate the percentage reduction in the number of elements processed by the diffusion model.
e) Explain how performing diffusion in latent space reduces computational requirements.

Q9. Discuss different approaches for accelerating diffusion model sampling. Explain:

- Denoising Diffusion Implicit Models (DDIM).
- Progressive distillation.
- Latent-space diffusion.

Compare their underlying principles and explain how these approaches address the computational challenges of conventional diffusion models.

Q10. Explain conditional diffusion models and how different types of conditioning information are incorporated into the U-Net architecture. Discuss an application of conditional diffusion models with the mechanisms used for Text conditioning and cross-attention.